

A Systematic Framework for Evaluating Retrieval-Augmented Generation Systems

Retrieval Quality, Groundedness, and Answer Usefulness

Abstract

Retrieval-augmented generation (RAG) combines information retrieval with large language models to improve responses on domain-specific or changing knowledge. However, evaluating a RAG system using answer fluency alone can hide retrieval failures and unsupported statements. This paper proposes a layered evaluation framework separating retrieval quality, evidence grounding, answer correctness, and user usefulness. The framework defines metrics and test procedures for each layer and discusses common failure modes such as retrieving irrelevant passages, incomplete evidence, and generating claims not supported by retrieved context. The paper argues that component-level evaluation is essential for diagnosing RAG systems and for comparing alternative retrieval pipelines.

Keywords: retrieval-augmented generation; large language models; information retrieval; evaluation; groundedness

1. Introduction

Large language models can generate coherent answers but may produce incorrect or unsupported information. Retrieval-augmented generation addresses part of this problem by supplying external passages to the model at generation time. Yet retrieval does not guarantee correctness: a system can retrieve the wrong document, retrieve insufficient evidence, or fail to use relevant evidence correctly.

This paper proposes an evaluation framework that treats RAG as a pipeline rather than a single model. The research objective is to identify where errors occur and to define measurements that distinguish retrieval failures from generation failures.

2. RAG Pipeline

A typical RAG pipeline contains document ingestion, chunking, embedding, indexing, query processing, retrieval, reranking, prompt construction, and answer generation. Each stage can introduce errors. Poor chunking may separate a definition from its context, while an inappropriate embedding model may retrieve semantically similar but factually irrelevant text.

Evaluation should therefore preserve intermediate artifacts such as retrieved passages and rankings. Without these artifacts, it becomes difficult to determine whether an incorrect answer originated from retrieval or generation.

3. Evaluation Framework

The proposed framework contains four layers. Retrieval quality measures whether relevant evidence appears among the top-k retrieved passages. Groundedness measures whether answer claims are supported by the retrieved evidence. Correctness measures agreement with a trusted reference answer or expert judgment. Usefulness considers whether the answer actually satisfies the user's information need.

A test set should contain both ordinary questions and difficult cases, including ambiguous queries, multi-hop questions, outdated documents, and questions for which the correct response is to acknowledge insufficient evidence. Such cases reveal whether a system knows when retrieval is inadequate.

4. Failure Analysis

One common failure mode occurs when the retriever returns passages that share vocabulary with the query but do not contain the needed fact. Another occurs when relevant evidence is retrieved but the language model ignores it. A third failure occurs when the evidence supports only part of a generated claim.

Separating these cases improves engineering decisions. Retriever changes should be judged primarily on retrieval metrics, while prompt or model changes should be judged on groundedness and answer quality. End-to-end evaluation remains necessary, but it should not replace component-level diagnosis.

5. Conclusion

RAG evaluation should move beyond subjective judgments of fluency. A layered framework covering retrieval, grounding, correctness, and usefulness can expose distinct failure modes and support systematic optimization. Future work should develop standardized datasets, automated claim-level evaluation, multilingual benchmarks, and evaluation methods for continuously changing knowledge bases.

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